

SIDE24 THEOREM PACKAGE v1.0

Artifact ID: GP-SIDE24-THM-001-v1.0

Date: 2026-08-02

Owner and operator: Dylan Roy

Track: three-dimensional SIDE24 compact-positive-mark theorem

Controlling disposition: RATIFIED-AT-STATED-SCOPE

Purpose: self-contained export surface for external human specialist review

1. Controlling statement

Let f be the normalized periodized Bargmann–Fock Gaussian field on the side-24 three-torus. Use the SIDE24 pair chart, Palm normalization, critical point typing, elder selection, and finite nonessential superlevel H_0 -lifetime convention fixed by the cited supplement. For pair separation r , pair direction t , level parameter b , and positive gap parameter κ , let $p_r(t, b, \kappa)$ be the conditional probability that the canonical compact-positive-mark configuration survives the collar, singular-near, far-witness, and elder-selection exclusions.

For every compact $K \subset \mathbb{R} \times (0, \infty)$, there are constants $C_K < \infty$ and $r_K > 0$ such that

$$\sup_{t \in S^2, (b, \kappa) \in K} (1 - p_r(t, b, \kappa)) \leq C_K r^3, 0 < r < r_K.$$

Consequently, the first-moment lifetime density $v_{3,24}$ of finite nonessential superlevel H_0 bars obeys

$$v_{3,24}(\ell) = c_{3,24} \ell^{-1/3} (1 + o(1)), \ell \downarrow 0.$$

The acceptance is exactly at this scope. It is not a statement about another torus side length, a different Gaussian law, essential bars, higher homology, an unbounded (b, κ) range, or the separate two-dimensional q0/P0.1 program.

The operator's signed controlling language is:

I, the human owner of this work, ratify the closure of RP-C and RP-S per the KIMI-AUD-006/006b audit chain and accept the theorem $\sup(1 - p_r) \leq Cr^3$ and the statement $v_{3,24}(\ell) = c_{3,24} \ell^{-1/3} (1 + o(1))$ at their stated scopes, with the carried dependencies recorded in AO48-AUD-044 §C.

The signed record is AO48-OPR-045-v1.0: Drive 1MfA94SaoYpnnAs7HYLNvowG9EHIR3Opk, 3,397 bytes, SHA-256 e48d7c271dff9c120316ee2129a303aed6f78b518c2e0c08337eb357bd60e2e8.

2. Constant and local parameterization

The two contact values are pinned as

$$f(M) = b, f(S) = b - \frac{\kappa r^3}{6}, \nabla f(M) = \nabla f(S) = 0,$$

with $M = x - rt/2$ and $S = x + rt/2$. Thus the height gap is $\ell = \kappa r^3/6$, and the gap-coordinate Jacobian contributes $r^3/6$. The pushforward $r = (6\ell/\kappa)^{1/3}$, together with the contact intensity and the $O(r^3)$ failure estimate, produces the exponent $-1/3$ in (T2).

Write $G_0^{all}(t, b, \kappa)$ for the limiting all-typed, determinant-weighted contact integrand after the pair blow-up and let $G_0^{all}(\kappa)$ denote its (t, b) -marginal over $S^2 \times \mathbb{R}$. The finite-side constant is represented by

$$c_{3,24} = \frac{6^{2/3}}{18} \int_0^\infty \kappa^{-2/3} G_0^{all}(\kappa) d\kappa.$$

Its infinite-volume reference value is

$$c_{3,\infty} = 2^{-23/6} 3^{-8/3} (29\sqrt{6} - 36) \Gamma(1/6) \pi^{-5/2},$$

numerically

$$c_{3,\infty} = 0.041775931840598343342936665428575556466681519661\dots$$

The certified side-24 correction is

$$\frac{c_{3,24}}{c_{3,\infty}} - 1 = -\frac{620813376}{35} e^{-288} + \varepsilon_{24}, |\varepsilon_{24}| < 10^{-180}.$$

The arithmetic route uses

$$P_3(L) = -\frac{L^2(10L^4 - 147L^2 + 315)}{105}, P_3(24) = -\frac{620813376}{35},$$

and the cone coefficient

$$D_2 = \frac{29}{6} - \sqrt{6}.$$

These identities belong to the constant chain. They do not replace the probabilistic and facewise estimates needed for (T1).

3. Proof architecture

The proof is organized as one local-to-global implication rather than as a collection of script outputs:

1. The contact Kac–Rice/Palm formula writes the small-lifetime first moment in the pair variables (x, t, r, b, κ) , with determinant weight and critical-point typing.
2. The Palm-repair normalizer satisfies $cr^2 \leq Z_r \leq Cr^2$.
3. RP-A/RP-L control the local same-pair failures, capture/escape events, and the weighted Palm/coarea passage by $O(r^3)$.

4. RP-F controls a fixed-distance or far witness by $O(r^3)$.
5. RP-C/RP-S control every collar and singular-near face, including the angular axis, by $O(r^3)$. This is the V3.4/FACEWISE closure.
6. The elder-selection recombination joins the mutually compatible local and far exclusions without changing the r^3 order.
7. The change of variables $\ell = \kappa r^3 / 6$, the full-mark integrable majorant of Theorem A.3.3, and the constant ledger give (T2) and (2.2)–(2.5). This is a genuine truncation-and-dominated-convergence argument on $S^2 \times \mathbb{R} \times (0, \infty)$, not an inference from compact convergence alone.

The role of each closing artifact is made explicit below.

4. Premise-by-premise proof-chain map

4.1 RP-A and RP-L: Palm/coarea and local same-pair control

Theorem G.7.1 supplies the local Palm/coarea interface and the same-pair failure estimate. Its proof is stratified into:

- covariance eigenfloors and Gaussian regression on the normalized contact variables;
- soft Hessian columns and determinant weights;
- the $Z_r \asymp r^2$ Palm normalizer and the required weighted moments;
- shallow value layers and the unique value-window integral of length $\kappa r^3 / 6$;
- Gaussian tail budgets;
- the exact gap parameter and the deterministic capture/escape geometry.

The conclusion used downstream is an $O(r^3)$ bound under the exact weighted Palm law, uniform at the stated compact-positive-mark scope. The closing carrier is the recovered V5 SIDE24_GAP_FILL_SUPPLEMENT.md: Drive 1D5eRshQWyiuglrlikBUhtasLyiyV1FX, 105,980 bytes, SHA-256 05967f14a8b8d6a9e2637210807872ba4f30810987cb7f3017e10b186d07f6da.

4.2 RP-F: fixed-distance endpoint and far-witness control

Section G.8 uses the same normalized pair pins and determinant-weighted Palm law to bound a witness outside the shrinking local charts. The joint density is nonsingular at fixed separation; its conditional mean/covariance and the Hessian moment envelope are uniform on the compact complement. The single value window contributes $O(r^3)$, while polynomial factors in $(|b| + \kappa)$ are controlled on the stated compact set. The result is the required $O(r^3)$ far-witness contribution.

G.8 is frozen in the same 105,980-byte supplement and therefore has the same Drive ID and SHA-256 as §4.1.

4.3 RP-C: collar faces

The collar proof does not extend a transverse compactness argument through the axis. It uses separate normalized charts:

Generic transverse face. With axial coordinate X and transverse radius ϱ , exact residualization and Fourier independence give

$$\det \text{Cov}(\nabla f(y) \mid V_r) \asymp \varrho^6 (4X^2 + \varrho^2)$$

on normalized strict-transverse compact charts, with uniform side-24 Schur constants.

Axial face. Exact Hermite divided differences factor the centered value and gradient. Positivity of the residual covariance is proved by Fourier independence. The exact axis has its own projective compactification; it is not obtained from (4.1) by compactness. The target mean yields an exponential penalty on the axial cone.

Midpoint face. The secondary expansion retains the fourth/fifth-order rows that survive the midpoint cancellation. Equivalently, the separated target-mean mismatch supplies an integrable exponential penalty. The full projective frame records the simultaneous (r, ρ, ζ) directions.

Endpoint faces. Reflected endpoint charts cover $q/r \rightarrow 0$ and $q/r \rightarrow 1$. Their exact Hermite residuals give the determinant and mean mismatch at the endpoint collision scale without borrowing a transverse constant.

Confluent row. In the s -scaled interpolating plane, put $\eta = r/s$, take the endpoint nodes at $X = \pm a$ with $a = \eta/2$, let c be the axial component of the normalized witness direction, and let σ be its signed noncollinearity coordinate (so the witness value-gradient minor is σ^6). FACEWISE v1.1 makes the third-order row explicit:

$$T_4 = \frac{3}{a^2} \left[\frac{p_X(a) + p_X(-a)}{2} - \frac{p(a) - p(-a)}{2a} \right],$$

whose unit limit is $p_{XXX}(0)$. This distinguishes the literal average/divided-difference normalization, whose ten-by-ten determinant is $2\eta^2\sigma^6$, from the confluent normalization, whose determinant is $-24\sigma^6$, independent of (η, c) .

4.4 RP-S: singular-near faces

The singular-near proof preserves both anisotropic quadratic factors. At the Euclidean contact face its covariance determinant has the form

$$\det C_{sing}^{BF} = \frac{\varrho^{10} (c^2 + \varrho^2) (3c^2 + \varrho^2)}{24}.$$

For the exact side-24 field the numerical factor is replaced by a continuous positive factor. A mixed axial blow-up is then used near the collinear face. The already established collinear target-mean mismatch produces the exponential penalty needed to absorb every fixed projective power.

The conditional three-Hessian moment bound is face-compatible. It retains the same monomial

$$D(r, s) = \left[r(r + s^2) \right]^2 (r^2 + s^3)$$

on the transverse, axial, endpoint, and singular-near faces. On noncollinear charts it is a pathwise polynomial-jet envelope; on the exact axis it is a density-weighted replacement. The false universal pointwise assertion is not used.

Here $s = |y - x|$ is the third-station radius, ϱ is the transverse distance from the pair axis, and $\Theta^2 = s^2 + \varrho^2$ on the singular-near chart. On the collar charts Θ denotes the corresponding projective scale ϵ_a or δ_m . The collar integrations split the near-axis and angular-dominant overlap:

- near the axis, $\Theta \asymp s$, and the exponential absorbs the s^{-6}/r^{-3} deficit;
- in the angular-dominant overlap, the deficit-free transverse estimate applies.

A single-chart pointwise absorption for $\Theta \gg s$ is not asserted. This explicit two-case display is the second FACEWISE v1.1 amendment.

The angular-axis integral has the uniform model bound

$$\int_0^{\rho_0} \rho (s^2 + \rho^2)^{-m/2} e^{-c/(s^2 + \rho^2)} d\rho \leq C_m.$$

After the value window, radial volume, determinant weights, and $Z_r^{-1} = O(r^{-2})$, the remaining monomials are

$$r^7 s^{-5}, 2r^6 s^{-3}, r^5 s^{-2}, r^5 s^{-1}, 2r^4, r^3 s^2,$$

with respective integrated orders

$$O(r^3), O(r^4), O(r^4), O(r^5 \log(1/r)), O(r^4), O(r^3).$$

Thus every collar and singular-near face, including the angular axis, contributes $O(r^3)$. In (4.5)–(4.7), $d\rho$ is the transverse radial measure after the angular variables have been separated, and the subsequent s -integration runs over the singular-near radial chart specified in FACEWISE Sections 5 and 9.

4.5 Elder selection and recombination

The canonical all-witness count uses one legitimate value integration $\int_{b-\kappa r^3/6}^b du$. The local collar, singular-near, and far regions form the prescribed cover; endpoint overlaps are counted through the reflected endpoint charts, and the elder-selection rule is applied to the same critical-point typing as the Palm formula. Union and first-moment bounds therefore preserve the common $O(r^3)$ order. No extra value-window factor is introduced, and no chart is promoted from a diagnostic grid alone.

The measurable elder mark, the disjoint elder-failure exhaustion, and the selected/all-typed recombination are carried by Parts A.1, F.1–F.2, and Theorem A.3.3 of the recovered 105,980-byte supplement, Drive 1D5eRshQWyiugIrlIkBUHltasLyiyV1FX, SHA-256 05967f14a8b8d6a9e2637210807872ba4f30810987cb7f3017e10b186d07f6da.

4.6 Full-mark integration and constant identification

The compact-positive-mark estimate (T1) supplies pointwise convergence of the selected integrand on each compact mark set, but the constant in (2.2) uses the whole mark space. Theorem A.3.3 closes that interface with the uniform majorant

$$C \left[1 + (|b| + \kappa)^N \right] e^{-c(b^2 + \kappa^2)} \kappa^{-2/3}.$$

This is integrable on $S^2 \times \mathbb{R} \times (0, \infty)$: the $\kappa^{-2/3}$ singularity is integrable at zero, while the Gaussian factor controls both noncompact tails. Truncate in (b, κ) , apply the compact facewise limit on each truncation, and then remove the truncation by (4.8). The off-diagonal lifetime density is uniformly $O(1)$, hence $o(\ell^{-1/3})$. The selected and all-typed limits agree because $0 \leq p_r \leq 1$, $p_r \rightarrow 1$, and the same majorant applies. Theorem A.3.3 and the exact constant recombination in Part B of the recovered supplement therefore close the global passage to (T2) and (2.2)–(2.7), under the same Drive ID and SHA-256 cited above.

5. Exact V3.4 and FACEWISE evidence

Role	Artifact	Drive ID	Byte s	SHA-256 / identity
Audited facewise input	rp_c_rp_s_facewise_closure_RELAY.m d	1rF9YAVw19fEG_QIRvHlesXev4n segA7w	35,31 0	40faad08824e4e77520143078ed606c8681532cc77ae 9c9d42551baa5277b400
Pre-audit disposition/check record	V3.4 closure disposition	1QJ_H5A- xg2TINB7CqHBo pjvY4-AVTJdJ	8,157	d262021d7c0819ba09a6680c29379e3d731e982705a 939bd60b8361e3382fbb1
Full audit-replay archive	SIDE24_V3_4_AU DIT_REPLAY_FUL L_ARCHIVE.zip	1TSH5V0f-7Bwdl HIY6- VarfuDFEQT-Tif	12,46 5,983	ba2a296b6a9c438eaf9f30aeaa40d4162195b256a9b2 516259cd30a2468dd480
Conditional-moment companion	three_hessian_con ditional_moment_e nvelope.md	1y0ZFI- S4BU7ZmBg7gD By7ts3J7oJDduu	12,14 5	f19132b47dd58339d632ffe8ef5c9466dbcf9b353b406a 31e22d3e35b5e5d0fb
Withdrawn-route audit note	quartic_remainder_ audit_note.md	17pL94ArcdpMG m89kwSGui9DnJ WnlH41I	5,830	e344ccecc3945ecd1c931e7e84eb25ef2fe2a0c9b41f57 68eb03619400b2b38db
Endpoint counterexample regression	verify_quartic_endp oint_counterexampl e.py	1EzOG80ErsPF78 5utTcBq7x_Dlp_ ZSPV1	13,32 9	461959368dfc7759c78ee105e4b82ff97659ddfc3a2883 7bbf7e8092da2f39d9
Post-006b authoring-conformance carrier	SIDE24_FACEWIS E_v1.1_SHA_d567 7359.md	1Yky0N- Y5M4HzST8- 1EFpgsvYPKksnY GL	36,54 2	d5677359e72243a89450c14d324051948d721aa031b b729f2ce9f76fb392804e

The full V3.4 archive is the source for the §6X withdrawal-completeness check; the shorter relay intentionally omits the withdrawn argument block.

FACEWISE v1.1 was authored after the 006b reconciliation and before AO48-OPR-045. Its d567... hash is therefore the post-audit authoring-conformance carrier covered by the later operator ratification. It must not be described as the byte object Kimi originally read.

6. Audit and operator chain

Stage	Record	Drive ID	Bytes	SHA-256 / qualification
Initial third-family audit reconciliation	AO48-AUD-043-v1.0	1wDuGvUxFnWTYe8PElgjo6upwzjdBMV4-	4,818	b41b1e1ff37632cd3034b4bcd73cc30b999745db2ef68b3ef28a3a1306ef0cc; records KIMI-AUD-006 only as 4b66f85e...5a38
APPROVE-conversion reconciliation and ratification package	AO48-AUD-044-v1.0	1qX_KJz2KreWDy7BXNmCBesUmbkJCoPBa	5,245	be6b53a21370984650733d9f00c60d0648b452fe65f32bf3ce93846f5c22737c
Operator ratification	AO48-OPR-045-v1.0	1MfA94SaoYpnnAs7HYLNvowG9EHIR3Opk	3,397	e48d7c271dff9c120316ee2129a303aed6f78b518c2e0c08337eb357bd60e2e8
Controlling status fold	GP-SIDE24-CTL-001-v1.0	1U8tFvVjqXjXO8Eih_SEUgy8AGBOJC_wp	5,117	ed688cb64d8ff5b6d8bb09d6d03df0668009ecbbff9bdc83e770a38a3cb86491
Native status mirror	SIDE24 post-ratification status fold	1my81FaIbVluMhzyxg34kXuBPCGOwJah_1abJv4mJWwo	native	mirrors the ed688cb6... controller
Native closure register	SIDE24 Closure Register R2.8	1Yjre5iYWJDHgxask_cuZoEFX4hXn9uCJUUnVFc478K9A	native	RP-C/RP-S/theorem state and carried dependencies

The exact raw texts of KIMI-AUD-006 and KIMI-AUD-006b have not yet been supplied to this Drive line. Their current corpus representations are the AO48-AUD-043 and AO48-AUD-044 reconciliation records and the operator's signed ratification. This package does not invent full Kimi hashes or quote text not present in those records. When Dylan Roy supplies the raw report texts, they should be landed as numbered, whole-file-frozen audit carriers and appended to this chain without changing the theorem statement.

7. Carried dependencies — verbatim

The following is reproduced verbatim from AO48-AUD-044 §C:

- (a) The frozen V3.3 eigenfloor tables — the $O(r^{-7})$ collar density origin and s^{-7} frame bounds — consistent with, but not re-derived in, the audit; same object class as Kimi's independently proved G.3/G.7.1 normalizer. (b) Three V3.4 diagnostic scripts absent from Kimi's corpus — non-blocking; every analytic claim they corroborate was independently recomputed. (c) The Palm-repair normalizer $cr^2 \leq Z_r \leq Cr^2$ (V3.3, previously audited).

Dependency (b)'s connected-Drive availability issue is now resolved; delivery of these carriers into Kimi's corpus remains pending. The three scripts are present byte-for-byte in Drive:

Script	Drive ID	Bytes	SHA-256	Replay
verify_hybrid_mixed_curvature_and_axis_absorption.py	1VPYlprjCWgNmoIMiVaTL5sa6wUirseVr	31,533	0ba99085cf5b1655deb95859a63f02478057491fe1d394124d853e93bea8c481	202/202 PASS; normal/optimized byte-identical
verify_facewise_collar_axis_integration.py	1a67DJHnF8NKPxDdhdKvIDpOHioFWCeQR	19,538	bb4fdb739636e504d2c2eeafd8d8582dbf1dba12225bca88788927b049d58e5f	62/62 PASS in both modes

Script	Drive ID	Byte s	SHA-256	Replay
diagnose_side24_gen eric_collar.py	1iylpIDkYR0YDi wv9vGRVQ- Rel8uh0sD	6,551	abc5b8cd3f03c251a8bf47115a9d0 097ef03ae62dd65debcb23082ba9 73176c0	ALL_CHECKS_PASS on its declared grid

Replay environment: Python 3.12.13; SymPy 1.14.0; mpmath 1.3.0; NumPy 2.4.4; SciPy 1.17.1. These scripts are regression and ledger evidence within their own scope statements. They do not replace the analytic Fourier-independence, compact-atlas, regression, conditional-moment, or Kac–Rice arguments.

The specific historical V3.3 eigenfloor-table carrier and its individual whole-file hash are not exposed by the current connected-Drive index. The tables are carried inside the full V3.4 replay archive, Drive 1TSH5V0f-7BwdlHIY6-VarfuDFEQT-Tlf, whose envelope is frozen at 12,465,983 bytes and SHA-256 ba2a296b6a9c438eaf9f30aeaa40d4162195b256a9b2516259cd30a2468dd480. No standalone table hash is invented here. The mathematical $cr^2 \leq Z_r \leq Cr^2$ normalizer is also stated in the historical V3.3 checkpoint; its independently citable proof is Theorem G.3.1 in the recovered 105,980-byte supplement, Drive 1D5eRshQWyiuglrlikBUhltasLyiyV1FX, SHA-256 05967f14a8b8d6a9e2637210807872ba4f30810987cb7f3017e10b186d07f6da. The absence of a separately indexed V3.3 checkpoint hash is a carrier-level limitation preserved by this package, not a new mathematical claim.

8. Failed approaches and correction ledger

The package preserves the unsuccessful routes because they define the proof's current safety boundaries.

1. **Mirror-sign error in the triple-contact display.** The printed $-det H_s$ cross-term originally had the wrong sign. After the third witness-value pin is restored, that sign controls the η^2 cancellation and is load-bearing. The corrected negative sign is used in V3 and later.
2. **Raw degree and collision overstatement.** The three-factor determinant product has generic degree six in κ , but the degree drops on coefficient-degeneracy faces $2c = \pm\eta$. Those faces are not geometric collisions unless the transverse coordinate also vanishes.
3. **Far-density order.** A claimed extra $O(\ell)$ density factor was withdrawn; the correct fixed-distance density is $O(1)$. The sole value window supplies the needed $O(r^3)$.
4. **Capture/escape threshold and tube cost.** An earlier deterministic threshold and a tube-cost count were insufficient. The G.7.1 route uses the corrected geometry and weighted Palm budgets.
5. **Unpinned midpoint subtraction.** A singular-near value row subtracted a random midpoint value not contained in the pin block. It was replaced by a pair-measurable residual and the s^{-7} normalized frame.
6. **Universal pointwise Hessian envelope.** The literal all-face pointwise curvature route fails on the exact axis. It was withdrawn and replaced by the hybrid monomial envelope: pathwise off-axis and density-weighted on-axis.

7. **Quartic endpoint remainder.** A purported uniform quartic remainder has an endpoint counterexample. The withdrawn §6X block is retained only as provenance; the live proof uses EP.1/EP.2 and the endpoint measure ledger.
8. **Confluent-row ambiguity.** The literal 10-by-10 average/divided-difference reading yields $2\eta^2\sigma^6$; it cannot be cited for the parameter-free coefficient. FACEWISE v1.1 writes the confluent row (4.2) explicitly.
9. **Single-chart axial absorption.** Pointwise absorption fails when $\Theta \gg s$. FACEWISE v1.1 displays the near-axis and angular-dominant regimes separately.
10. **Independent-audit overreach.** Early local checks supported only the components they exercised; they did not close RP-A/RP-L, RP-C/RP-S, RP-F, or elder recombination. The final status rests on the complete facewise proof, the third-family audit/reconciliation chain, and operator ratification.
11. **Fail-open reproduction scripts.** Bare Python assert statements and scripts that printed FAIL while returning status zero were treated as non-evidentiary for closure. Later verifiers use explicit fail-closed checks and optimized-mode transcript comparisons.
12. **G.9.1 pin-block factorization.** The historical polynomial factorization of the pin determinant was false. The corrected planar determinant was independently reconstructed as $(P + gU)(2gP^2 + 2vUP + g vU^2)$, where $q = e^{-d^2/2}$, $U = u_1^2$, $P = u_2^2 + u_3^2$, $g = \text{Var}(f_{xy} \mid \text{pins})$, and $v = \text{Var}(f_{xx} \mid \text{pins})$. The explicitly labeled reconstruction and its 26/26 verifier are indexed by the SIDE24 G.9/R2 Recovery Index, Drive 1yQQfid0m0U7Wwu0_e0zr663hrAflZy0srRZGkT0tPyg. That recovery is supporting local covariance evidence, not a substitute for the facewise side-24 theorem.

9. Reopening rule

The ratified result reopens only upon one of the following:

1. an exact counterexample to an audited display;
2. failure of a carried V3.3 eigenfloor table; or
3. a landed diagnostic script contradicting a claim it was said to corroborate.

An unavailable raw report, a stale historical HOLD banner, or a diagnostic script's absence from a reviewer's corpus is a preservation or routing issue, not by itself a mathematical counterexample. Such issues must nevertheless be recorded honestly, as the Kimi raw-carrier gap is recorded here.

10. Track firewall

Nothing in this package changes the q0/P0.1 program. GP-DER-118 and LS-CTL-003-v1.1 remain controlling there; PZ0/N1/E0/E2/D0/I0 and the independence lanes remain open, LB-RATE remains measured-grade, and P0.1 remains HOLD. No q0 sealing, release, or cross-track promotion is authorized. The blind $d^3/48$ midpoint-reflection result remains frozen and untouched pending the pre-registered Kimi Task-6 comparison.

11. External-review route

An external specialist can review the result in this order:

1. verify the operator scope and status in AO48-OPR-045 and this package's separate freeze receipt (a file cannot contain its own whole-file hash);
2. read G.7.1 and G.8 in the recovered 105,980-byte supplement;
3. read the audited 35,310-byte V3.4 facewise input and the 12,145-byte three-Hessian envelope;
4. inspect FACEWISE v1.1 only for the two authoring-conformance amendments, using the provenance distinction in §5;
5. audit the §6X withdrawal against the full V3.4 archive and the quartic counterexample pair;
6. replay the three diagnostic carriers and the endpoint counterexample under the pinned environment;
7. inspect AO48-AUD-043, AO48-AUD-044, and AO48-OPR-045; then append the direct KIMI-AUD-006/006b carriers when supplied;
8. keep the three carried dependencies in §7 visible when assessing the theorem's stated scope.

This is an export and routing surface. The frozen mathematical sources remain the authoritative proof bodies.

12. Package-local controllers

- Raw post-ratification status fold: Drive 1U8tFvVjqXjXO8Eih_SEUgy8AGBOJC_wp, SHA-256 ed688cb64d8ff5b6d8bb09d6d03df0668009ecbbff9bdc83e770a38a3cb86491.
- Native status mirror: Drive 1my81FalbVluMhzyxg34kXuBPCGOwJah_1abJv4mJWwo.
- Diagnostic handoff: Drive 1dolk9RcGWt7vRNopLmbfDXZi0yv8upjl, SHA-256 6aef6bd213876245e74040700370c8dc632fab9efef9296143739a8b55c0984e.
- Raw closure register R2.8: Drive 1isOnrX0VG9QeLOTAIrinArHPyzZZu2Sa, SHA-256 fbfa7f8f095a82e9a8a0c62401d73dd6101252db93411149a788ac6b4751e43f.
- Native closure register R2.8: Drive 1Yjre5iYWJDHgxask_cuZoEFX4hXn9uCJUnVFc478K9A.
- Coupled research registers: Drive 1O6x8ivmaVUxYqKmCOXmToIHpMXqDI362ibqBI8HY8no; SIDE24 append rows are recorded without changing any q0 row.

End of SIDE24 THEOREM PACKAGE v1.0.